

SIMATS ENGINEERING
Slot: C, Course Code: ITA0501, Course Name: Computer Vision

Duration: 60 mins

Activity Tool 3: Class Test (Low)

Marks: 20

Assessment Tool Weightage: 10%

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Q. No.	Question
1	An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.
2	A dark image lacks detail. Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.
3	An image has poor contrast. Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.
4	A low-contrast image requires uniform intensity distribution. Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.
5	An image shows uneven brightness across regions. Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.
6	A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.
7	An image requires noise reduction without significant loss of details. Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.
8	An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.
9	Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.
10	A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Code of Conduct:

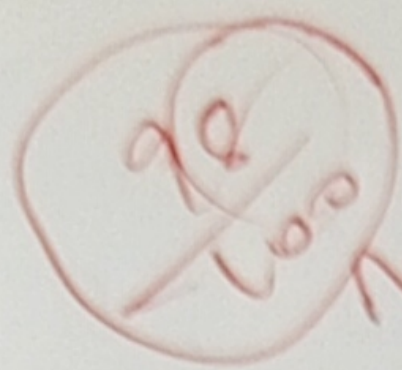
I SOMISETTY VENKATA SAI NAVEEN KUMAR (192425176) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student
SOMISETTY VENKATA SAI
NAVEEN KUMAR
192425176

RUBRICS FOR CALCULATION

Criteria	<i>Understanding of Concepts</i>	<i>Explanation</i>	<i>Application</i>	<i>Organization</i>	<i>Accuracy</i>	<i>Clarity</i>	Total
Weightage	25%	25%	20%	15%	10%	5%	100%

Signature of the Course Faculty



CO2-AT3

Class Test

S. Naveen Kumar

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ITAD501 - Computer Vision.

4/8/26

- 1) Image appears too Bright.
Use Negative Transformation. It converts bright pixels into dark pixels using $s = 255 - r$, making hidden details more visible and improving image clarity.
- 2) Dark image lacks detail.
Use Gamma Transformation with $\gamma < 1$. It increases the brightness of dark regions and enhances low intensity detail.
- 3) Poor Contrast image
Use Histogram Equalization. It spreads intensity values uniformly across the histogram, improving the overall contrast.
- 4) Low-contrast image needs uniform intensity.
Apply Histogram Equalization. It spreads intensity values uniformly across the histogram, improving

the overall contrast.

- 5) Uneven brightness across regions.
Use Adaptive Histogram Equalization.
It enhances contrast locally and corrects non-uniform illumination without over-enhancing noise.

- 6) Random noise in image.

Use the Mean filter. It replaces each pixel with the Average of its neighbouring pixels, reducing random noise.

- 7) Noise reduction without losing details

Use the Median filter. It removes salt and pepper noise while preserving edges and ~~reducing~~ restoring fine details lost during ~~smoothing~~ important details.

Image blurred after smoothing

Use unsharp Masking. It sharpens the image by enhancing edges and restoring fine details lost during smoothing.

fine details and not clearly visible.

Use the Laplacian filter. It enhances edges and high frequency components making fine details more visible.

10)

Noisy image before analysis.

Use the Gaussian filter. It smooths the image by reducing Gaussian noise while preserving the overall image structure for further processing.